

Virtual Connections: Fostering Social Interaction and Creativity for Marginalized Populations Through XR

Ali Adjorlu*

Aalborg University

Anton Buus Hansen

Aalborg University

Aske Lau Kaiser

Aalborg University

Catharina Riis-Jensen

Aalborg University

Lasse Viking Kristensen

Aalborg University

Marius Langemark Arnbjerg

Aalborg University

Nor Hady

Aalborg University

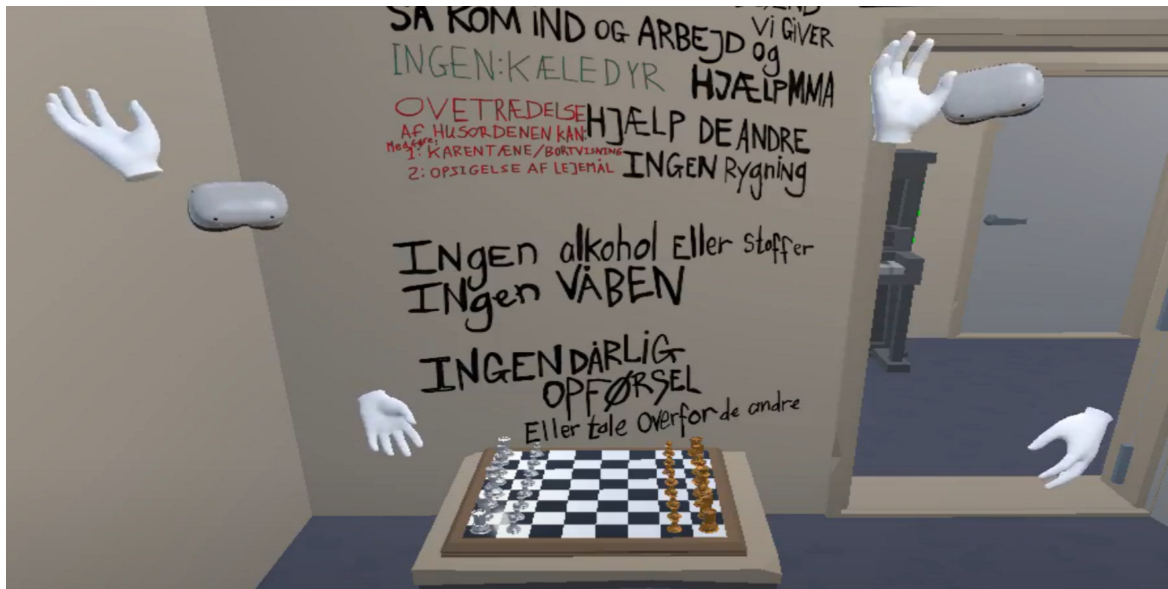


Figure 1: Two players interacting in the chess room

ABSTRACT

ocial inclusion remains a critical challenge for individuals with severe mental illness (SMI), many of whom face significant psychological and geographical barriers to accessing supportive environments. This position paper presents the design, development, and evaluation of a virtual reality (VR) application replicating the social and creative spaces of *Maskine Maskine Amager* (MMA), a Danish recovery institution known for its peer-based approach to mental health support. The VR application offers an immersive platform for individuals unable to visit MMA physically, allowing them to visit a virtual version of the space, engage in creative activities, interact with other MMA volunteers and visitors, and experience a sense of community from the safety of their homes.

The VR experience was designed through an iterative, user-centered approach, involving MMA stakeholders throughout the process to ensure alignment with the institution's ethos. An explorative study at the MMA presented in this position paper advocates that VR holds the potential to bridge the gap between virtual and physical participation, which can encourage gradual transitions to in-person engagement and social inclusion.

Index Terms: Virtual Reality, Social Inclusion, Mental Health, Virtual Communities

*e-mail: adj@create.aau.dk

1 INTRODUCTION

Maskine Maskine Amager (MMA) ¹ is a Danish recovery institution that provides a unique peer-based approach to supporting individuals with Severe Mental Illness (SMI). Unlike traditional psychiatric facilities, MMA fosters a sense of community and empowerment through creative and social activities, allowing participants, referred to as "machinists," to engage on their own terms. MMA's philosophy revolves around mutual support, with the mantra, "By helping MMA and the other machinists, you are also helping yourself." This approach integrates art, music, and social interaction as tools for recovery while destigmatizing mental illness [1].

Despite its success, MMA faces challenges in reaching individuals who are unable to physically access the institution due to geographical distance or mental health barriers such as agoraphobia or severe anxiety. Virtual reality (VR) offers an innovative solution to these challenges, enabling the replication of MMA's physical and social spaces in a virtual environment. This allows potential users to engage with MMA's unique creative and social atmosphere remotely.

VR has emerged as a powerful tool for creating immersive social environments that transcend physical boundaries [2, 7]. These technologies have shown potential for fostering social interaction among marginalized groups who face geographical or psychological barriers to accessing physical spaces [4, 8].

Research has demonstrated that VR interventions can improve social skills and reduce symptoms of social anxiety among individ-

¹<https://mmamager.dk>

uals with mental health conditions [10, 9]. By simulating supportive and inclusive environments like MMA, VR can serve as a stepping stone, encouraging individuals to transition to physical spaces for further social integration [3]. Additionally, VR's customization capabilities empower users to interact at their own pace and comfort level, addressing barriers to engagement for marginalized populations [5, 6].

In this paper, we present the design and evaluation of a VR application that emulates the social and creative environment of MMA. The goal is to explore how virtual spaces can promote social interaction among individuals with SMI and encourage them to engage with physical community spaces. Our findings contribute to the growing body of research on XR technologies as tools for social inclusion and mental health recovery.

2 METHODS

2.1 Iterative and User-Centered Design Approach

The VR application was developed using an iterative, user-centered design approach aimed at replicating the social and creative atmosphere of MMA. The process spanned three iterations, each involving feedback from MMA stakeholders and subsequent refinements to address identified challenges.

2.1.1 First Iteration: Initial Prototype

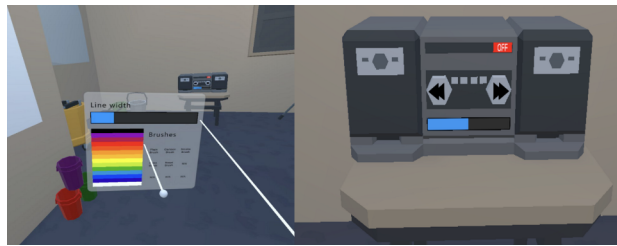


Figure 2: Screenshot of the MMA gallery as represented in the virtual environment.

The first iteration focused on creating a foundational prototype to explore the potential of virtual environments in replicating MMA's spaces. Using the game engine Unity 3D, a low-poly representation of key areas of MMA was created, including the painting room and social areas. Interactive features such as basic 3D painting tool (see figure 2) and multiplayer functionality were included to demonstrate the concept while the users could navigate the environment using continuous locomotion.

This prototype was evaluated through a showcase to two of MMA volunteers experienced in running the creative workshops for machinists. Feedback highlighted the need for greater fidelity to MMA's physical environment, emphasizing the inclusion of symbolic elements such as artwork and furnishings. Additionally, they would like to have interactive elements in the application, such as the ability to play chess and draw and they pointed out the absence of voice communication for multiplayer sessions.

2.1.2 Second Iteration

Based on the feedback on the primary prototype, the second iteration introduced more refined and interactive features. Key updates included a more detailed gallery space displaying artwork from MMA (see figure 4) a music room with virtual instruments (interactive drum and piano). Additional activities, such as playing a virtual piano, drums, and chess (see figure 1), and a bike workshop where the users could work together to assemble and repair virtual bicycles were integrated to promote creativity and collaboration. All these are activities that are available in the physical

MMA. The users avatars are presented using a generic HMD (see figure 1).

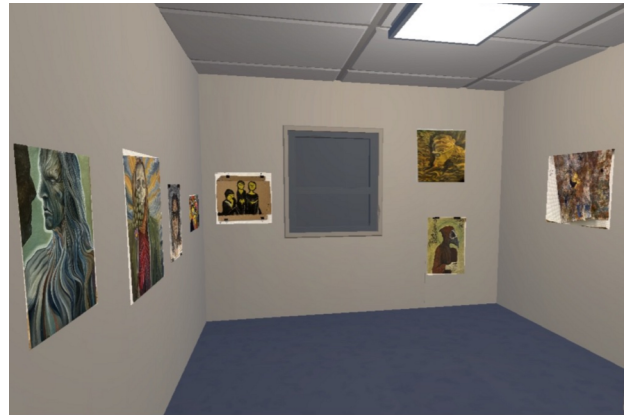


Figure 3: Screenshot of the MMA gallery as represented in the virtual environment.

This version was evaluated during a second showcase at MMA with machinists and two volunteering staff. Observations and informal discussions revealed improvements in the representation of MMA's atmosphere but also highlighted areas for further enhancement. Users noted the importance of additional environmental details, such as furnishings and symbolic decor, to better reflect MMA's unique identity. Interaction mechanics, particularly object manipulation, were identified as requiring refinement. Moreover, onboarding processes for new users needed further simplification to accommodate those unfamiliar with VR.

The feedback from this evaluation guided subsequent redesign efforts. Decorative elements and props were added to enhance environmental fidelity, interaction mechanics were optimized for smoother usability, and onboarding instructions were improved to ease the learning curve for new users.

2.1.3 Final Iteration

The third and final iteration incorporated all previous feedback to deliver a polished application that faithfully represented MMA's creative and social spaces. This version included comprehensive representations of MMA's key areas 3, enhanced interaction mechanics, and additional features to support social engagement, such as real-time voice communication and collaborative experiences such as a radio in the 3D painting room where the users could listen to music together while painting.

2.2 Final Iteration: Explorative Evaluation Session

The evaluation of the final iteration was conducted through an *explorative evaluation session* held at the MMA on four machinists.

The session began with each machinist entering the VR environment and freely exploring its features, including 3D painting tools, multiplayer interactions, and creative activities such as playing virtual instruments. The MMA volunteer joined each machinist in the virtual space, acting as a guide and interaction partner to facilitate engagement. This one-on-one setup allowed the machinists to explore the application at their own pace while benefiting from real-time assistance and encouragement from the volunteer.

Following the completion of all individual sessions, the four machinists participated in a group discussion to reflect on their experiences. This discussion was intentionally informal, creating a relaxed atmosphere where participants could share their impressions without pressure. The group format encouraged a collaborative exchange of ideas, with machinists building on each other's observa-



Figure 4: Top: The floor plan of the MMA. Bottom: The virtual MMA seen from above

tions and providing collective feedback on the VR application. The informal nature of the discussion was key to maintaining comfort and openness among participants, many of whom had limited experience with VR.

3 RESULTS

The findings from the explorative evaluation are divided into four categories: user experience, usability, multiplayer interactions, and environmental representation.

3.1 User Experience

Participants generally expressed positive experiences during the VR session. Several users noted that the application provided a sense of immersion and engagement, with comments such as “you lose the sense of time and space.” The painting tools and interactive features, such as the music room, were well-received, with users describing them as enjoyable and stimulating. However, one participant reported minor motion sickness, particularly during navigation.

3.2 Usability

The usability of the application received mixed feedback. While participants adapted to the controls after some initial exploration, challenges were reported in certain areas. Picking up objects, such as chess pieces or musical instruments, was occasionally described as difficult. Navigation using snap-turn controls was functional but prompted requests for smooth-turn options. Additionally, participants found some interactions, such as painting, less intuitive than expected, with the ray-cast interaction system identified as a source of confusion.

3.3 Multiplayer Interactions

The multiplayer functionality successfully encouraged meaningful social interactions among participants, reflecting the communal ethos of MMA. These interactions occurred in various forms, ranging from verbal communication to collaborative activities and non-verbal gestures, demonstrating the potential of the VR application to foster a sense of community and engagement.

Collaboration was a key aspect of the observed interactions. Participants often engaged in shared activities, such as playing chess or

exploring the virtual environment together. For instance, one participant suggested exploring the kitchen collaboratively, showcasing how the application could replicate the cooperative experiences of the physical MMA environment.

Verbal communication played a significant role in enhancing social presence within the virtual space. The integrated voice chat allowed participants to converse naturally, exchanging greetings and coordinating activities. In one instance, a participant remarked, “Yes, now I can see you,” upon acknowledging another avatar entering their field of view. These exchanges highlighted the importance of real-time communication in establishing connections and maintaining engagement during collaborative activities.

Non-verbal interactions also contributed to the social dynamics of the VR environment. Gestures such as waving were used to acknowledge one another, demonstrating how even simple non-verbal cues could enhance the sense of presence and interaction in the virtual setting.

The involvement of MMA volunteers enriched these interactions further. Acting as guides, the volunteers provided support and facilitated engagement by helping participants navigate the virtual environment and discover its features. Their role mirrored their responsibilities in the physical MMA space, reinforcing their supportive presence and ensuring that all participants felt included and comfortable.

Additionally, the shared use of virtual spaces, such as the painting room and social areas, created opportunities for participants to connect and interact in ways that mirrored real-world communal settings. Comments like “It is the togetherness and presence with others” illustrated how the platform successfully replicated the inclusive and social atmosphere of MMA.

These observed social interactions demonstrate the effectiveness of the multiplayer functionality in fostering community and collaboration. They also provide valuable insights into how future iterations of the application could further enhance the social experience, such as through more dynamic non-verbal communication options or additional collaborative features.

3.4 Environmental Representation

The virtual environment was recognized as a faithful representation of MMA’s physical spaces. Participants noted that the painting room, gallery, and social areas captured the essence of MMA, with comments highlighting the authenticity of the decor and spatial layout. Suggestions for improvement included adding more decorative elements, such as rugs and additional props, to enhance the immersive quality further.

3.5 Additional Observations

Participants provided various suggestions for future enhancements. These included expanding the range of interactive activities, such as adding more instruments or allowing users to manipulate objects like artwork within the environment. Other ideas included smoother locomotion options, proximity-based voice chat, and the ability to use the virtual space for practical purposes like planning events.

The results highlight the application’s ability to engage users and foster social interaction, while also identifying areas for improvement in usability and environmental fidelity.

4 DISCUSSION

The results of this study highlight the potential of VR applications to increase accessibility to institutions like MMA for individuals who are unable to attend in person due to mental disabilities. By replicating MMA’s social and creative spaces in a virtual environment, VR provides a safe and inclusive platform for individuals facing psychological or geographical barriers. Users can explore

the virtual version of MMA, interact with volunteers and other machinists, and express themselves creatively—all from the safety and comfort of their homes.

For individuals who may be too mentally ill to leave their homes, the application can potentially offer a crucial opportunity to engage in social and creative activities that are otherwise inaccessible. The presence of MMA volunteers in the virtual space provides real-time support and guidance, ensuring that participants feel included and understood. Multiplayer interactions, such as collaborative painting or exploring the virtual environment together, further reinforce a sense of community and social connection, addressing the isolation often experienced by individuals with severe mental illnesses.

This virtual engagement can also serve as a stepping stone for users to build confidence and trust, potentially bridging the gap to eventual physical attendance at MMA. By allowing users to familiarize themselves with MMA's spaces, activities, and community ethos in a safe and controlled manner, the application has the potential to reduce anxiety and uncertainty about visiting the institution in person. This could encourage a gradual transition to physical engagement, supporting their overall recovery journey.

Currently, the application does not assign specific roles to MMA volunteers and machinists in the virtual environment. This design choice was made to maintain equality and inclusivity, allowing all participants to engage on the same level. However, it may be beneficial to explore the impact of distinguishing MMA volunteers with identifiable avatars. Providing volunteers with distinct visual representations, such as unique clothing or badges, could make their supportive role more apparent and foster a greater sense of structure in the virtual environment. Future iterations of the application could investigate the effects of such visual distinctions on the user experience, including whether they enhance or hinder the feeling of community and collaboration.

Future iterations of the VR application will also focus on enhancing accessibility and personalization to better meet the needs of diverse users. Planned features include options to choose between locomotion techniques, such as continuous locomotion or teleportation, to reduce the risk of cybersickness. Additional interaction mechanics, such as customizable pick-and-place functionalities and adjustable control schemes, will be developed to improve usability and inclusivity for individuals with varying levels of VR experience or physical ability.

Moreover, the application's adaptability and scalability suggest its broader potential for similar recovery centers or institutions with a focus on peer-based support. As a tool for social inclusion, it can extend the reach of these institutions, enabling them to support individuals who may otherwise remain isolated.

In summary, the VR application presented in this explorative paper represents an innovative solution for making places like MMA more accessible. By promoting meaningful social interaction, reducing anxiety through familiarization, and enabling creative expression, it offers a unique avenue for individuals to engage with supportive communities. This could pave the way for deeper integration into physical spaces and enhance their recovery outcomes.

ACKNOWLEDGMENTS

The authors would like to express their gratitude to Søren Dixen, Head of the Board at MMA, for facilitating this collaboration. Special thanks are extended to the volunteers and machinists at MMA who actively participated in the user-centered design process, providing invaluable insights and feedback that shaped the development of the application.

REFERENCES

- [1] M. M. Amager. Mma's recovery philosophy and creative spaces, 2023. Accessed: 2023-12-19. 1

- [2] D. Freeman, S. Reeve, A. Robinson, et al. Virtual reality in the assessment, understanding, and treatment of mental health disorders. *Psychological Medicine*, 47(14):2393–2400, 2017. 1
- [3] C. N. Geraets, E. C. van der Stouwe, R. Pot-Kolder, and W. Veling. Advances in immersive virtual reality interventions for mental disorders: A new reality? *Current Opinion in Psychology*, 41:40–45, 2021. 2
- [4] O. D. Kothgassner, A. Felnhöfer, N. Hauk, E. E. Kastenhofer, J. Gomm, I. Kryspin-Exner, and L. M. Glenk. Virtual social support buffers stress response: An experimental study. *PLoS One*, 11(1):e0147845, 2016. 1
- [5] F. Pallavicini, A. Ferrari, and F. Mantovani. Video games for the prevention and treatment of anxiety disorders: A systematic review. *Games for Health Journal*, 7(2):85–99, 2018. 2
- [6] K. Park, J.-K. Ku, S. H. Choi, H. J. Jang, J. Y. Park, S.-I. Kim, and J.-J. Kim. A virtual reality application in role-plays of social skills training for schizophrenia: A randomized, controlled trial. *Psychiatry Research*, 189(2):166–172, 2011. 2
- [7] G. Riva, R. M. Baños, C. Botella, B. K. Wiederhold, and A. Gaggioli. Positive technology: Using interactive technologies to promote positive functioning. *Cyberpsychology, Behavior, and Social Networking*, 19(2):69–77, 2016. 1
- [8] M. Rus-Calafell, P. Garety, E. Sason, T. J. Craig, and L. R. Valmaggia. Virtual reality in the assessment and treatment of psychosis: A systematic review of its utility, acceptability and effectiveness. *Psychological Medicine*, 48(3):362–391, 2018. 1
- [9] J. Turner, N. Ralph, and D. Shotton. Improving clinical outcomes in social anxiety disorder through the use of virtual environments. *Frontiers in Psychiatry*, 5:88, 2014. 2
- [10] L. R. Valmaggia, L. Latif, M. J. Kempton, and M. Rus-Calafell. Virtual reality in the psychological treatment for mental health problems: An systematic review of recent evidence. *Psychiatry Research*, 236:189–195, 2016. 2